Predicting Compressive Strength and Optimal Replacement Level for Recycled-Concrete-Fines Blended Cement from a Sparse Literature Dataset

August 21, 2026

Abstract

We build a machine-learning pipeline to predict the compressive strength and the optimal cement-replacement level of concrete blended with **recycled concrete fines / recycled concrete powder** (RCF/RCP), from a dataset compiled from **104 research papers**. As with the companion rice-husk-ash corpus, the binding problem is data representation rather than model choice: strength values are packed inside free-text cells as *(replacement level* $\times$ *curing age)* grids, spread across two differently-shaped sheets. Parsing and merging them recovers **1825 data points across 85 papers**, of which 1284 also resolve a curing age. A six-axis ablation shows that (i) absolute strength across papers is unpredictable ($R^2 = -0.18$) but becomes learnable once the paper's own control strength is supplied; (ii) unlike the smaller RHA corpus, *added features help here*—chemistry and canonicalised categoricals both improve generalization at 44 papers where they overfit at 23; and (iii) text embeddings still provide no reliable gain. A tuned, monotonically-constrained gradient-boosted model reaches $R^2 = 0.448 \pm 0.175$**, MAE** $= 9.9$ **MPa**. That number should be read against a hard **noise ceiling of** $R^2 = 0.80$ which we quantify: 180 of the 260 (paper, level, age) cells contain several distinct mixes whose activation treatment the dataset does not record. Finally, the optimal-replacement target turns out to be ill-posed rather than merely hard—a paper's reported optimum and the argmax of its *own* strength table disagree by 14 percentage points at $r = 0.43$, and for 62% of papers the strongest mix measured is the 0% control.

1 Introduction

Recycled concrete fines (RCF), also reported as recycled concrete powder (RCP), are the fine fraction left over when waste concrete is crushed. Using them to replace part of the Portland cement in a new mix diverts a low-value waste stream and cuts the CO_2 footprint of the binder. The two engineering quantities of interest are the **compressive strength** of the resulting concrete and the **optimal replacement level**. Our goal is a predictive model for both, trained on a dataset hand/LLM-extracted from the literature.

This report deliberately mirrors the methodology of our earlier study on rice husk ash (RHA), so that the two materials can be compared on equal terms. Section 9 draws that comparison out; the short version is that RCF has roughly twice the usable data and behaves quite differently as a material.

2 The Dataset: What It Actually Contains

The source workbook `Master_sheet_RCF.xlsx` has **two sheets**, and that is its most important structural property.

`Sheet1` holds 225 columns over 105 data rows under a **four-row hierarchical header**: super-banner (→ PAPER METADATA/INPUT/OUTPUT), category, feature (55 features), and a sub-field. Every feature is split into four columns—`Extracted Value`, `Unit`, `Source Type`, `Exact Evidence`—of which only the first two carry predictive information. The first five columns are single-level metadata (title, authors, journal, year, DOI), so $5 + 55 \times 4 = 225$. One paper appears twice under different capitalisation, leaving **104 unique papers**.

`Experimental Matrix` is a flat six-column, 1162-row table that is *already exploded to one row per mix*, carrying `Mix ID`, `Replacement Level`, `Activation Method`, `Key Processing Parameters` and `Reported Strength`. It covers 103 of the 104 papers and is joined on title.

Two traps are specific to this workbook and both are silent failure modes:

- **Missing values are the literal string** `"NULL"`, very often with a trailing parenthetical—`NULL (Note: the paper reports water/cement (w/c) ratio...)`. An equality test against `"NULL"` keeps those strings as data.
- **Units are inconsistent within a single column.** `Specific Surface Area` alone appears as m^2/kg, m^2/g and cm^2/g; `Bulk Density` as kg/m^3 and g/cm^3; the strength column includes one paper reporting psi and one reporting kg/cm^2. Values must be converted *before* any plausibility filter, or a range guard silently deletes two thirds of a column.

Among the 104 papers, 93 report a compressive strength, 92 declare their replacement levels, and 66 report an optimum. **Two papers are excluded as out of scope** (logged to `excluded_papers.csv`): both replace *aggregate* rather than cement, so their "replacement level" does not mean the same thing as everyone else's, and pooling them would reintroduce exactly the confound that makes framing A unlearnable.

Sparsity is uneven rather than uniform. Among the 85 papers that yield strength data, availability of the 30 retained features ranges from 100% (material family) and 97% (processing method) down to 8% (BET surface) and 6% (amorphous content)—that is, **0–94% missing, with 18 of the 30 features more than half missing** (Figure 1).

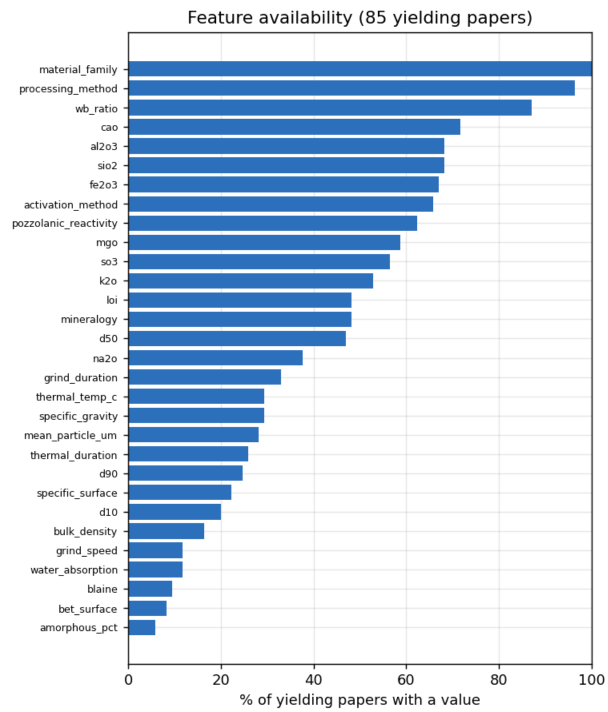


Figure 1: Per-feature availability among the 85 yielding papers. Material family, processing method and w/b are well populated; fineness measures (BET, Blaine, specific surface) and amorphous content are scarce. The dataset is larger than the RHA corpus but no less sparse.

3 Data Representation: Parsing Two Sheets Into One Table

Each `Sheet1` strength cell encodes a *grid* of measurements—and sometimes a third w/b axis—in one of ∼16 textual layouts, for example

```
Ref (0%):  / 2 days:  26.6 ± 0.89 / 28 days:  45.8 ± 1.45 ...
w/b 0.4:  / 0% RP: ~48.5 / 15% RP: ~47.3 ...
3-Day:  33.2 (MA20), 18.1 (MA50), 3.4 (MA100) ...
```

We built a line-oriented, context-carrying parser (`02_parse_explode.py`) that resolves, for each value, a replacement level and—where stated—a curing age, from an inline qualifier, a segment header, or persistent line context. It reads *both* sheets into one schema and de-duplicates on the measurement itself. Precision safeguards include:

- unsigned number matching, since a leading `-` is always a separator in this corpus;
- **reading the strength only from the value zone of an item.** This is not a detail. In `R20:  48.1` and `N-0.50-0:  28.5` the mix label contains numbers, and treating the item as a flat string makes the parser report strengths of 20 and 0.50 MPa. Items are first split into `label:  value` or `value (label)`; two such bugs cost roughly 0.2 of R^2 until they were found by hand-checking parsed rows against raw cells;
- a **per-paper whitelist**: a resolved level must match one of the paper's declared `Cement Replacement Level(s)` (plus 0 for the control), which rejects mis-read mix IDs such as `MFP400` and `Series 7`;

- unit conversion (psi $\times 0.00689476$, kg/cm^2 $\times 0.0980665$) and physical clamps ($0 \leq \% \leq 100$, $0 < \text{age} \leq 400\,\text{d}$, $0.5 \leq \text{MPa} \leq 250$).

This recovers **1825 data points across 85 of the 102 in-scope papers**—976 from the `Sheet1` grids and 849 from the Experimental Matrix (Figures 2 and 3). **1284 points over 57 papers** also resolve a curing age, 384 of them at the standard 28-day age. The 445 logged parse failures are overwhelmingly cells that report only a *relative* change (`3-day: +18%, 28-day: +16%`), a bare number list, or a figure reference, and are genuinely unrecoverable without the source PDFs.

Curing age is not assumed. Only 458 of the 1162 matrix rows state an age. Treating the rest as 28-day measurements would be fabrication, so they are retained with `age_known = False` and excluded from the main modelling table; Study 6 then tests that assumption empirically rather than baking it in.

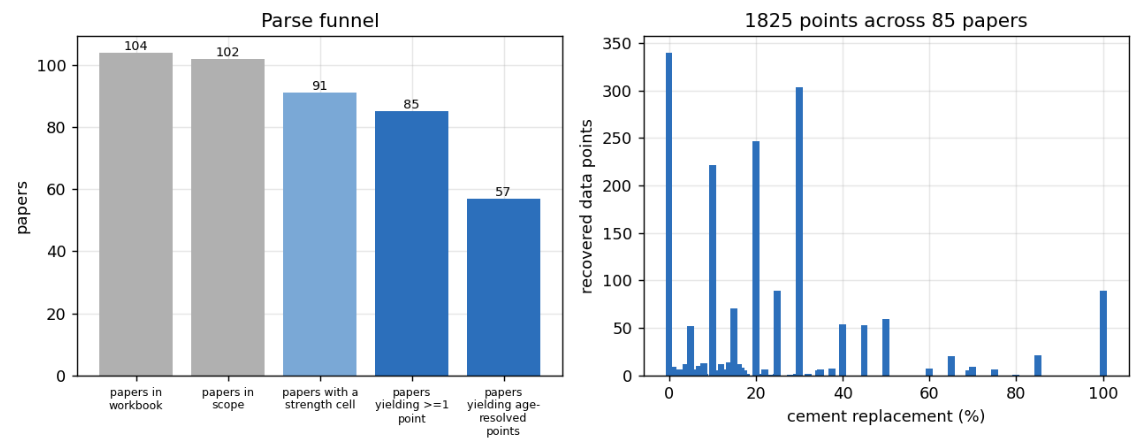


Figure 2: Parse funnel (left) and the distribution of recovered points by replacement level (right). The corpus is concentrated at 10–30% replacement, with a long tail out to 100%.

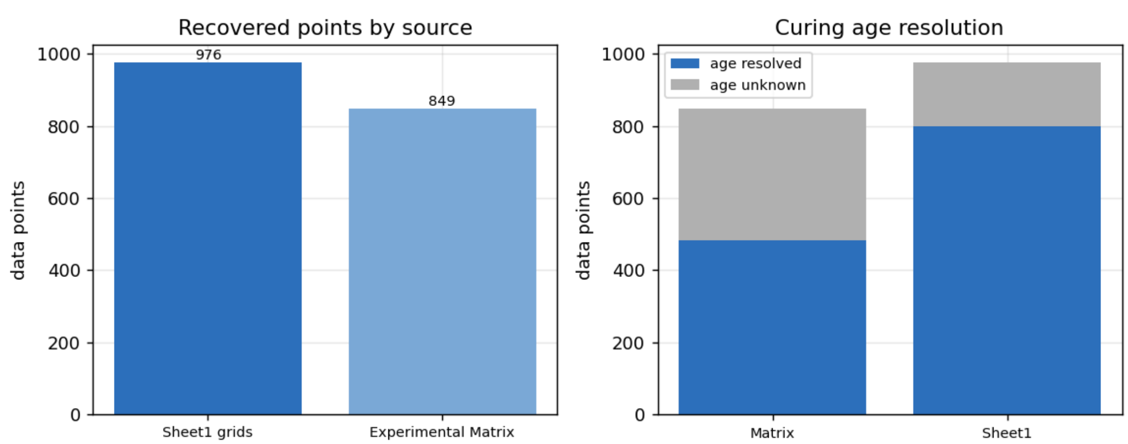


Figure 3: The two sheets contribute comparable numbers of points (976 and 849), but differ in what they resolve: 800 of the 976 `Sheet1` points (82%) state a curing age, against 484 of the 849 matrix points (57%). Both sources are needed—Study 6 shows that training on the matrix alone costs more than half the achievable R^2.

Checks against expected physics. Both checks must be made *within* a paper: a raw cross-paper average is dominated by mix type—the parsed corpus spans 0.6 MPa pastes to a 184 MPa UHPC, and 0.6–108 MPa among the age-resolved rows these panels use—rather than by the variable of interest. Normalising each *(paper, replacement level)* series by its own 28-day value, strength does rise with curing age: 77% of pre-28-day points fall below their own series' 28-day value and 83% of later points exceed it, with the median ratio climbing 0.62 (2 d) → 0.83 (7 d) → 1.00 (28 d) → 1.12 (90 d). The one-day point is an exception, sitting at 0.87 rather than below the two-day value; it is drawn from a different subset of series (only some papers report a one-day strength at all) and is a composition effect rather than a reversal of the trend.

The strength-versus-replacement shape is where RCF differs most from RHA. Of the 31 papers reporting at least three distinct levels at 28 d, only 7 (23%) peak at an interior level, while **22 (71%) are highest at the lowest level tested** and 2 (6%) are still rising at the highest. For this material, replacing cement usually just costs strength—consistent with recycled concrete powder being far less pozzolanic than rice husk ash and acting largely as a filler. That fact drives the negative result in Section 8.

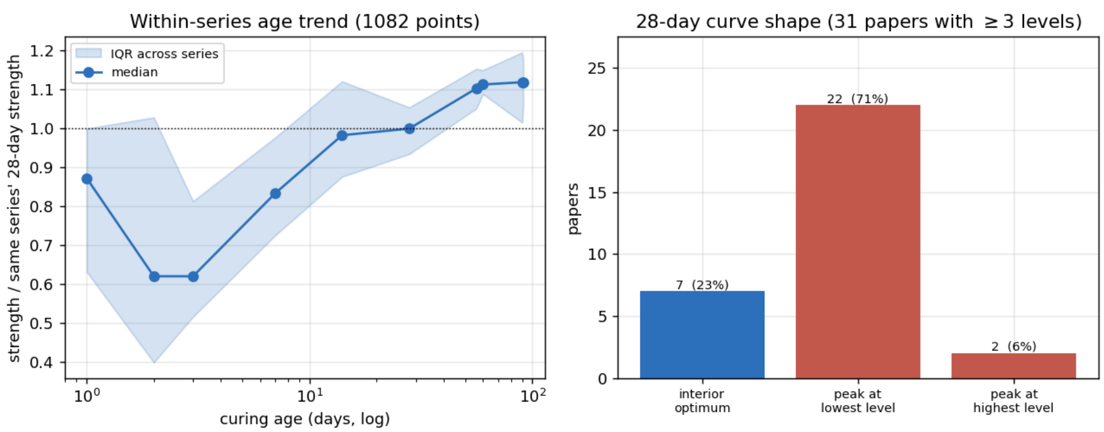


Figure 4: Physics checks on the parsed data, computed within papers. **Left:** each *(paper, replacement level)* series is normalised by its own 28-day strength; the median rises with age (shaded band is the IQR across series). The 1-day point breaks the pattern because a different, smaller set of series reports it. **Right:** the 28-day strength-vs-replacement curve has an interior optimum in only 23% of papers; in 71% the best level tested is the lowest one, frequently the 0% control.

4 Feature Engineering

Numeric cells are parsed as the mean of the numbers they contain, but only after **unit normalisation** and inside a **physical range guard**. The guard is load-bearing rather than cosmetic: the w/b cell `RC: 0.41 / C30R: 0.45 / C100R: 0.48` contains 30 and 100 from the mix labels, and an unguarded mean reports a water/binder ratio of 26.

High-cardinality free text is **canonicalized** into compact categoricals: activation method ∈ {none, thermal, *carbonation*, mechanical, chemical, multiple, other}—carbonation is given its own level rather than being folded into a generic "other", because CO_2 curing is a central RCF-specific treatment; processing method ∈ {crushing, milling, thermal, combined, other}; material family ∈ {powder, fines, paste, cdw, aac, other}, which folds the 90 free-text `Material Type` spellings together; and pozzolanic reactivity as an ordinal, with negative markers tested first so that "limited pozzolanic reactivity" does not score as high. We deliberately apply **no imputation** for the tree models, which handle missing values natively; imputers are evaluated separately in Study 4 and only ever fit *inside* each cross-validation split.

5 Methodology

Because rows from the same paper are strongly correlated, all evaluation uses **grouped cross-validation by paper**: no paper appears in both training and test. Given the sample size, a single split is unreliable, so we report the mean and standard deviation of **repeated grouped hold-out** (`GroupShuffleSplit`, 30% of papers held out, 40 repeats). We compare three target framings:

A. Absolute strength, all papers predict MPa directly from RCF features.

B. Relative-to-control predict strength normalized by the paper's 0% control.

C. Absolute strength given control predict the RCF-blend strength using the paper's control (0%) strength at the same age as an additional input—always known in practice, hence not leakage. Control rows are excluded from evaluation to avoid a trivial identity.

Framing C is evaluated on **823 blend rows over 44 papers**.

A noise ceiling worth stating up front. 180 of the 260 *(paper, replacement level, age)* cells in framing C contain *more than one mix*—different activation treatments, and sometimes different w/b ratios, at the same coordinate. Their spread is irreducible for any model that cannot observe the treatment, and decomposing the variance puts a hard ceiling of $R^2 = 0.80$ on this task as posed. Every R^2 below should be read against 0.80, not against 1.0.

6 Ablation Study

We ablate six axes. All numbers are repeated grouped hold-out (mean $\pm$ std) over 40 repeats, except four rows noted with † where 36–39 repeats survived (a split is discarded when a fold ends up with no usable rows). Because those rows are averaged over an easier subset of splits, *their RMSE is not comparable* to the 40-repeat rows; comparisons within a study are otherwise like-for-like.

Study 1 — Target framing. Absolute strength across papers is *unpredictable* ($R^2 = -0.184$): the corpus mixes pastes, mortars and normal concrete, spanning **0.6–108 MPa** over the 1284 age-resolved rows framing A is scored on, and those baseline levels are set by mix factors absent from the data. (The 184 MPa corpus maximum quoted in Section 3 is *not* in this set—that paper states no curing age, so it never enters an age-dependent framing.) Supplying the control strength (framing C) makes the RCF effect learnable (Table 1).

Table 1: Study 1: target framing (XGBoost, all structured features).

Framing	R^2	std	RMSE	MAE
C: blend | control strength	0.341	0.207	14.18	10.77
B: relative-to-control	−0.170	0.234	0.32	0.24
A: absolute, all papers	−0.184	0.320	20.30	16.05

Study 2 — Model. The tree ensembles are tightly bunched and all clearly beat the mean baseline; HistGradientBoosting leads on the mean, with ExtraTrees, Random Forest, GradientBoosting and XGBoost within 0.04 of it—well inside the run-to-run spread of $\sigma \approx 0.17$–0.22, so this ordering is not by itself reliable (Table 2). What *is* decisive is the failure of the non-tree candidates: Ridge ($R^2 = -23.1$) and a small MLP ($R^2 = -229$) are catastrophic on this tiny, collinear, missing-laden matrix, and the stacked ensemble is dragged below zero by its Ridge component. Section 7 therefore searches over the two gradient-boosted candidates rather than picking a winner here.

Study 3 — Feature set. This is where RCF departs from the RHA corpus. There, the three-feature base set won outright and *every* added group hurt. Here, **added features help**: the canonicalised categoricals lift R^2 from 0.311 to 0.373 and chemistry to 0.350 (Table 3, Figure 5a). With 44 papers rather than 23, the model can afford them. Text embeddings still cannot be justified—MiniLM and TF-IDF both land *below* the plain base set.

Table 2: Study 2: model (framing C, all features).

Model	R^2	RMSE
HistGB†	0.384	13.67
ExtraTrees	0.371	13.89
GradientBoosting	0.353	14.00
Random Forest	0.353	13.98
XGBoost	0.341	14.18
Stack (RF+XGB+Ridge)	−0.028	16.87
Mean baseline	−0.080	18.42
Ridge	−23.1	59.69
MLP (64,32)	−229.2	167.55

†39 of 40 repeats; RMSE not strictly comparable.

Table 3: Study 3: feature set (framing C, XGBoost).

Feature set	R^2	RMSE
base + categorical	0.373	13.80
base + fineness	0.370	13.84
base + all numeric	0.356	13.98
base + chemistry	0.350	14.06
all structured	0.341	14.18
all + MiniLM text	0.313	14.39
base [repl, age, control]	0.311	14.48
all + TF-IDF text	0.310	14.46
base + processing	0.272	14.77
base + mix descriptors	0.268	14.90

Study 4 — Imputation. No imputer beats native NaN handling on like-for-like terms. MICE posts the highest mean ($R^2 = 0.362$) but completed only 36 of 40 repeats, and median and KNN completed 39 while scoring *below* native; since none improves R^2 reliably and each adds a fitted component that must be re-fit inside every split, we keep **native handling** (Table 4).

Study 5 — Physics priors. Adding **monotone constraints**—strength non-decreasing in curing age and in control strength—gives a consistent gain, $0.311 \rightarrow 0.339$ (Table 5). Unlike the RHA study, this is not an XGBoost-only capability: scikit-learn's `HistGradientBoostingRegressor` supports `monotonic_cst` as well, so the prior does not constrain the model choice.

Table 4: Study 4: imputation (framing C).

Scheme	R^2	RMSE
XGB + MICE†	0.362	13.87
XGB + native	0.341	14.18
XGB + median†	0.298	14.59
XGB + KNN†	0.297	14.59
Ridge + median	−23.1	59.69

†36–39 of 40 repeats; RMSE not comparable to the native row.

Table 5: Study 5: physics priors (base features).

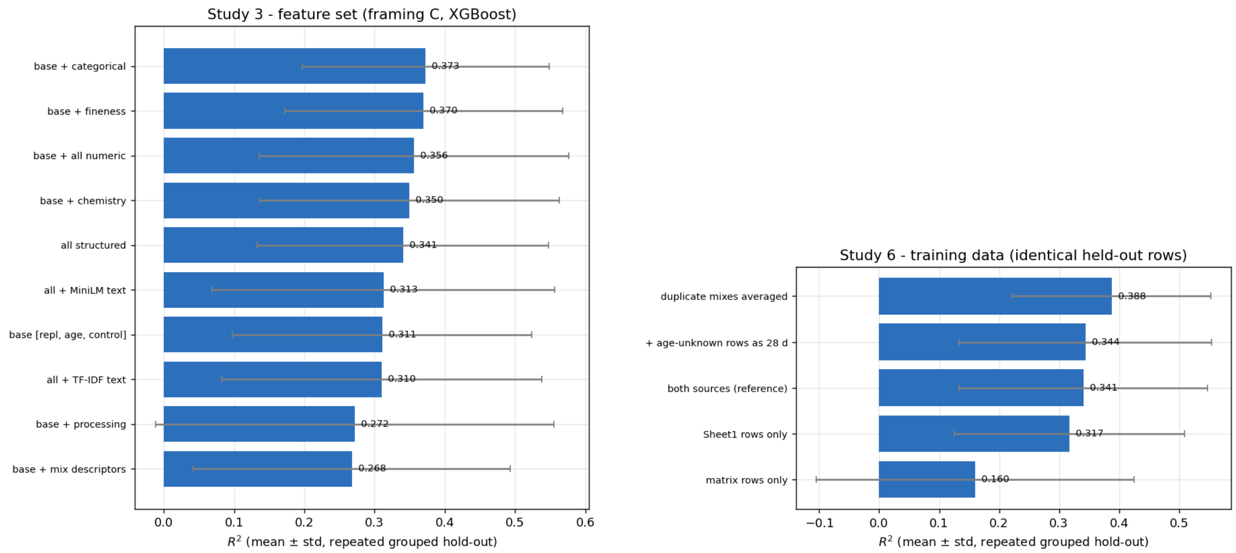
Prior	R^2	RMSE
monotone(age, control)	0.339	14.10
log + monotone	0.327	14.17
log-target	0.311	14.44
plain	0.311	14.48

Study 6 — Which rows to train on. This axis has no counterpart in the RHA study; it exists because the RCF workbook has two sources and a large pool of rows whose curing age never resolved. Here *only the training rows vary*—the held-out rows are identical in every variant—so these R^2 values are directly comparable to one another, which they would not be if the evaluation set changed too. Three results (Table 6):

- **Averaging the duplicate mixes** in training is the single largest gain in the study, $0.341 \rightarrow 0.388$. This is the noise ceiling of Section 5 showing up as a modelling choice: when several treatments share one coordinate, their mean is a better training target than the individual points.
- **Both sources are needed.** Training on the Experimental Matrix alone collapses to 0.160; the `Sheet1` grids alone reach 0.317; together, 0.341. The two halves are of similar size (380 and 443 of the 823 rows, covering 30 and 34 of the 44 papers), so this is not simply a row-count effect: the matrix half is far more *redundant*, its 380 rows occupying only 100 distinct (paper, level, age) coordinates against the grids' 201, because the matrix lists many mixes per coordinate.
- **Assuming 28 d for the age-unknown rows is neutral** ($0.341 \rightarrow 0.344$, well within $\sigma = 0.21$). The assumption buys nothing, so we do not make it.

Table 6: Study 6: training data (framing C, XGBoost; held-out rows identical across variants).

Training set	R^2	std	RMSE
duplicate mixes averaged	0.388	0.166	13.69
+ age-unknown rows as 28 d	0.344	0.210	14.14
both sources (reference)	0.341	0.207	14.18
`Sheet1` rows only	0.317	0.192	14.42
matrix rows only	0.160	0.265	15.98



(a) Study 3 — feature set.

(b) Study 6 — training data.

Figure 5: The two ablation axes where RCF behaves differently from RHA. Bars are mean ± std over 40 repeated grouped hold-outs.

7 Final Model and Results

Rather than hand-pick the ablation winners, we searched the architecture space they opened up: {HistGB, XGBoost} × five feature sets × {average duplicate mixes in training or not} × {monotone or not}, 40 configurations each scored over the same 40 repeated grouped hold-outs (`architecture_search.csv`). Averaged over everything else, each axis moves R^2 by a small but consistent amount: features `base+cat+chem` 0.410 vs `base` 0.355; monotone 0.388 vs unconstrained 0.378; averaging duplicates 0.386 vs not 0.379; and the two model families are indistinguishable (HistGB 0.383, XGBoost 0.382). The best single configuration was XGBoost on `base + categorical + chemistry`, training on averaged duplicates, with monotone constraints. A 108-point hyper-parameter grid over that configuration then selected a shallow, heavily-regularized model (`max_depth`=2, `learning_rate`=0.1, 500 trees, `min_child_weight`=5, `reg_lambda`=3.0).

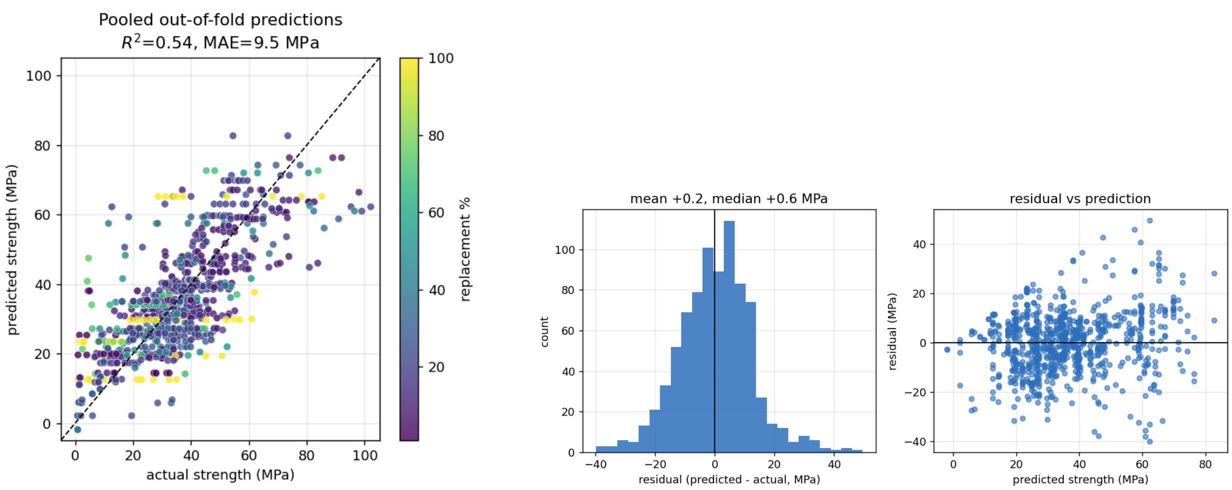
	R^2	RMSE (MPa)	MAE (MPa)
Final model (repeated grouped hold-out)	$\mathbf{0.448 \pm 0.175}$	**12.9**	**9.9**
Best ablation single-axis result (Study 6)	0.388 ± 0.166	13.7	10.4
Mean baseline	−0.080	18.4	14.3
Noise ceiling (Section 5)	*0.80*	—	—

The model therefore captures a little over half of the variance that is available to be captured. That framing matters: 0.45 against a ceiling of 0.80 is a substantially better result than 0.45 against 1.0 would be, and it says the remaining headroom is mostly in *recording the mix treatments*, not in fitting harder.

Figure 6a shows out-of-fold predictions. Its panel title reports $R^2 = 0.54$, MAE $= 9.5$ MPa, which is *not* the headline number above: the scatter comes from a single 5-fold `GroupKFold` pass whose out-of-fold predictions are pooled into one R^2 over all 823 rows, whereas the table reports the mean of 40 grouped hold-outs that each train on fewer papers (70% vs 80%) and are scored split-by-split. The pooled single-pass figure is the more optimistic of the two; we quote the repeated hold-out as the honest estimate and show the pooled predictions only because they give one prediction per row to plot.

Residuals (Figure 6b) are close to centred—mean +0.2 MPa, median +0.6 MPa—with no systematic bias at ordinary strengths, but the tails run from −40 to +50 MPa. The large negative tail is the familiar high-end failure: the five worst rows are all above 80 MPa and all under-predicted by 35–40 MPa, and they come from just three papers: two mixes of paper 78 (102 and 97 MPa), one of paper 103 (99 MPa), and two of paper 71 (84 and 81 MPa). Only one paper in the framing-C set exceeds 100 MPa, so under grouped validation that paper is always wholly in the test fold and the model has never seen an example of its regime when it is scored.

Feature importance (Figure 7) is dominated by the paper's control strength (0.48 of the total). What is striking is how little the two other base inputs carry: curing age ranks 6th at 0.037 and **replacement level ranks only 9th at 0.021**, below several oxides. The chemistry block collectively takes 0.27 and the canonicalised categoricals 0.20. Read together with Study 3—where adding those blocks was what improved generalization—this says the model is largely predicting *which kind of study this is* from its material description, and then anchoring on the control strength. The replacement level itself is a weak predictor, which is the same fact the curve-shape analysis of Section 3 reported from the other direction: for RCF, moving the replacement level mostly just moves strength down along an already-known trend.



(a) Out-of-fold predictions.

(b) Residual diagnostics.

Figure 6: Final model behaviour, from a single pooled 5-fold `GroupKFold` pass ($R^2 = 0.54$ pooled, versus the 0.448 ± 0.175 repeated-hold-out estimate quoted in the table above). Colour encodes replacement level. The dashed line is the ideal $y = x$; the points falling far below it at the right are the three high-strength papers.

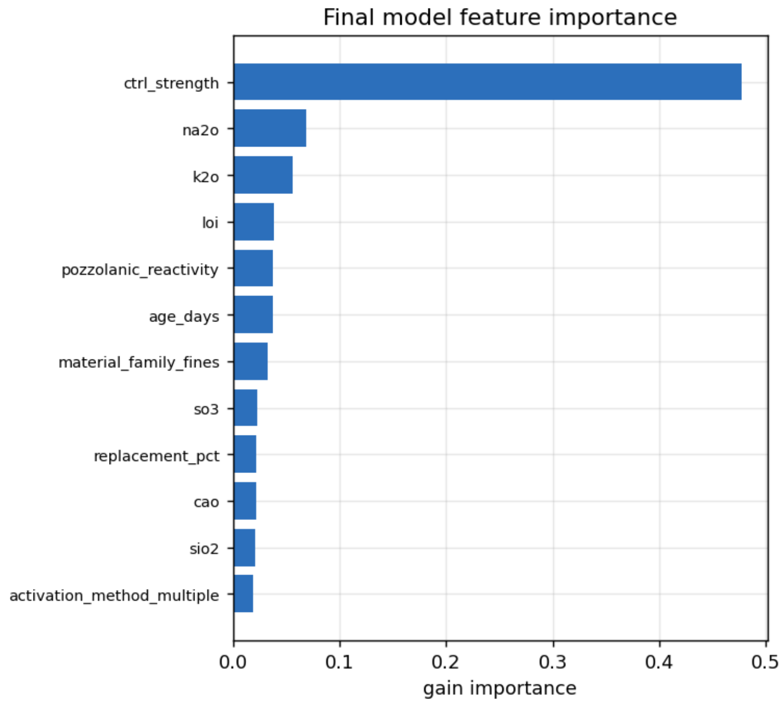


Figure 7: Feature importance of the final model (top 12 of 37 columns after one-hot encoding). The paper's control strength dominates at 0.48; replacement level does not appear in the top eight.

8 Predicting the Optimal Replacement Level

We derive the optimum as the replacement level maximizing the predicted 28-day strength curve of the final model, swept over 0–50% in 2.5-point steps and evaluated on held-out papers. It does not beat a naive prior: on the 32 held-out papers with a reported optimum, the argmax approach gives MAE = 14.0 percentage points against a mean-baseline of 8.6.

Unlike the RHA study, the failure is *not* degeneracy. That model collapsed to a single constant for every paper; this one produces five distinct values spanning 0–45% (standard deviation 6.8), and its predictions are concentrated where the strength data says they should be—9 papers at 0%, 21 at 5%, 11 at 7.5%. The model is behaving sensibly. The problem is the label.

The target is measuring two different things. Each paper supplies two candidate "optima": the one it *reports in prose*, and the argmax of its *own tabulated 28-day strengths*. On the 29 papers that provide both, these disagree by **MAE** = 14.1 **percentage points** with a correlation of only $r = 0.43$ (Figure 8, right). The reported optima average 18.8%; the empirical argmax of the same papers' own data averages 5.4%, and for **18 of the 29 papers (62%) the empirical optimum is** 0% —their control mix is the strongest thing they measured.

The two labels therefore disagree with each other *more* than our model disagrees with either of them (14.0 against the reported optimum, 7.8 against the empirical one). This is not noise to be modelled away. A paper that recommends "20% RCP" while its own table shows strength falling monotonically from 0% is not making an arithmetic error—it is optimizing something else: cost, CO_2 saving, workability, or an acceptable strength loss relative to a target grade. None of those criteria are recorded in this dataset.

Neither label is beatable here, for different reasons. Against the reported optimum, the target encodes objectives the features do not contain. Against the empirical optimum, the task is close to degenerate in the other direction: since 62% of papers peak at 0%, the constant predictor "always 0%" already achieves MAE = 5.05 over the 39 papers with an empirical optimum, better than our model's 7.81 and exactly equal to the median prior. Predicting the strength-maximizing replacement level for RCF is largely the trivial answer "do not replace any cement," and the interesting question—how much cement can be replaced for an acceptable strength penalty—is not the question this label asks.

We therefore report the optimum task as a negative result with a diagnosis rather than a number to improve: the corpus needs a defined optimality criterion before the target is well-posed.

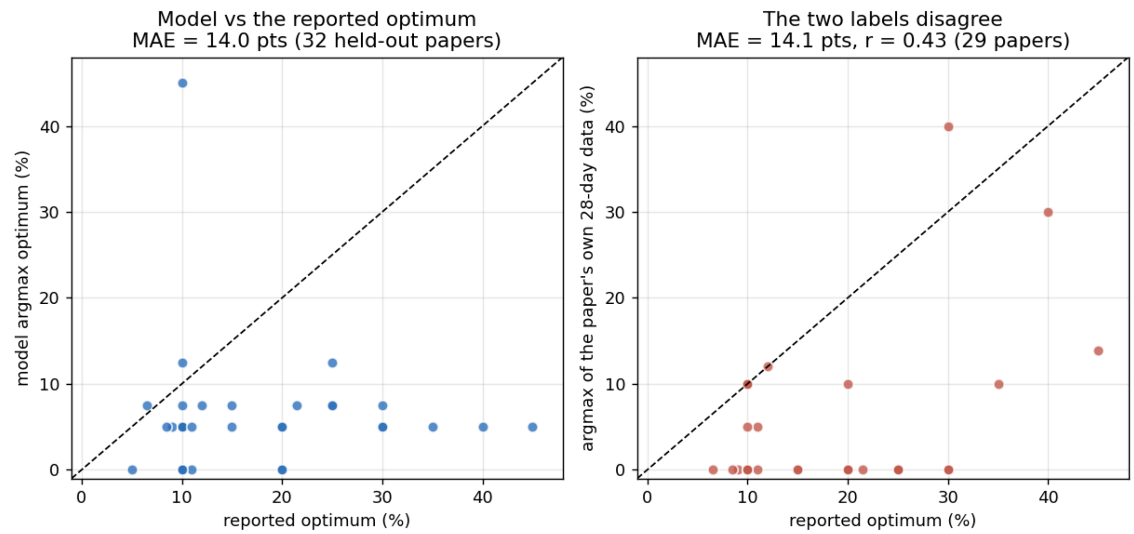


Figure 8: Left: the model's argmax against each paper's reported optimum; MAE = 14.0 points over 32 held-out papers, worse than the 8.6-point mean prior. **Right:** why. Each paper's reported optimum plotted against the argmax of its *own* 28-day strength table—the two disagree by 14.1 points at $r = 0.43$, with the reported values systematically higher. The label is not measuring strength maximization.

9 Comparison with the Rice-Husk-Ash Corpus

The two workbooks share a schema and were processed with deliberately parallel pipelines, so the differences in Table 7 are differences in the data and the material, not in method.

Table 7: RHA versus RCF, from the two pipelines' own outputs.

	RHA	RCF
Papers in the workbook	79	104
Recovered strength points	756	1825
Papers yielding points	44	85
Papers with a 0% control (age-resolved)	25	49
Framing-C rows / papers	335 / 23	823 / 44
Papers peaking at an interior level	42%	23%
Does adding features help?	no (overfits)	yes
Final model R^2 (repeated grouped hold-out)	0.50 ± 0.20	0.448 ± 0.175
Final model MAE (MPa)	10.1	9.9

The two final models land in the same place on MAE and are within noise of each other on R^2; the RCF model has the tighter spread, as one would expect from twice the papers. But the two R^2 values are not measuring against the same ceiling—the RCF task carries the 0.80 noise ceiling of Section 5 because its duplicate mixes are unresolvable, a limit the RHA corpus does not have—so the apparent tie understates the RCF model.

Three further points are worth drawing out.

RCF has roughly twice the usable data, largely because the workbook ships a pre-exploded Experimental Matrix alongside the packed grids. That extra sample size is what flips the Study 3 result: feature groups that overfit at 23 papers earn their place at 44.

The materials behave differently. An interior strength optimum is common in RHA (42% of papers) but the exception in RCF (23%), where 71% of papers are strongest at the lowest replacement level tested. Rice husk ash is a genuinely pozzolanic material that can improve strength at modest replacement; recycled concrete powder mostly dilutes the binder. Any framing that presumes an interior optimum is therefore a worse fit for RCF than for RHA. Both optimum-prediction studies fail, but for different reasons, and the distinction matters: the RHA model failed *mechanically*, collapsing to a constant 7.5% because its depth-2 monotone structure could not move an argmax; the RCF model behaves sensibly and fails because the *label* is ill-posed (Section 8). Only the second diagnosis points at the data collection rather than the model.

Both corpora are limited by unrecorded mix descriptors, in different ways. In RHA the missing information is the base mix (cement grade, aggregate, superplasticizer), which is why absolute strength is unpredictable across papers. In RCF that problem persists *and* is joined by a second one: the activation treatment varies within a single (paper, level, age) coordinate and is recorded only as free text, which is what produces the $R^2 = 0.80$ ceiling of Section 5. Structuring the per-mix activation and w/b columns is the highest-value next step for this dataset—higher than any further modelling work.